

popquiz levers, machines, equilibrium - bonus

 This is a preview of the published version of the quiz

Started: Apr 29 at 8pm

Quiz Instructions

bonus points

!! equilibrium means $\text{mass1} \times d1 = \text{mass2} \times d2$

A crow bar is a lever of class 1. amplifies distance in this case.

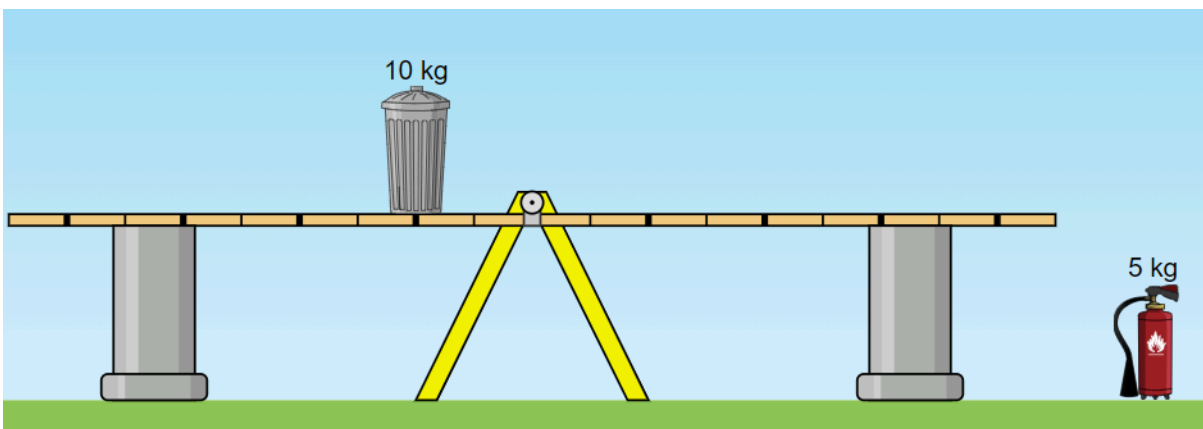


Question 1 15 pts

https://phet.colorado.edu/sims/html/balancing-act/latest/balancing-act_en.html 

(https://phet.colorado.edu/sims/html/balancing-act/latest/balancing-act_en.html)

You use this simulation to help you.



10kg is placed at the position 2 units of distance from the fulcrum. Where should we placed the 5 kg on the otherside? You can verify with the App by removing the poles.



2 units from the fulcrum



6 units from the fulcrum



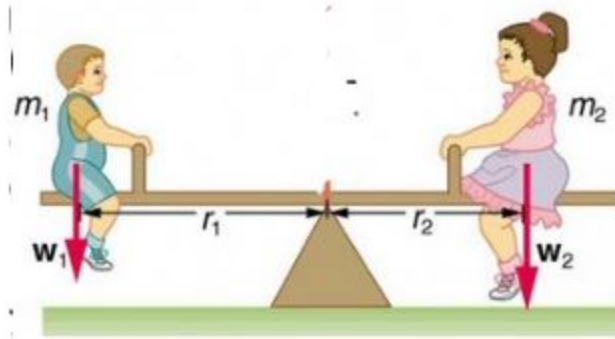
impossible to balance out



4 units from the fulcrum



Question 2 14.5 pts



Little boy has a mass of 100 pounds and sit at one end of the Seesaw. Little girls has a mass of 150pounds. The Seesaw has a length of 8 meters (!!! the total length is 8m so little boys is m from the fulcrum). How far from the fulcrum is little girl?

you can verify here:

https://phet.colorado.edu/sims/html/balancing-act/latest/balancing-act_en.html

(https://phet.colorado.edu/sims/html/balancing-act/latest/balancing-act_en.html)

select balancing act. Use 10 kg for little boy and 15 kg for little girl (same ratio as 100lbs and 150lbs). It will not balanced perfectly because the simulation does not have decimals. The distances are 1, 2, 3 ...



5,3m (about 5m_



6m



2.7m (about 3m)



4m

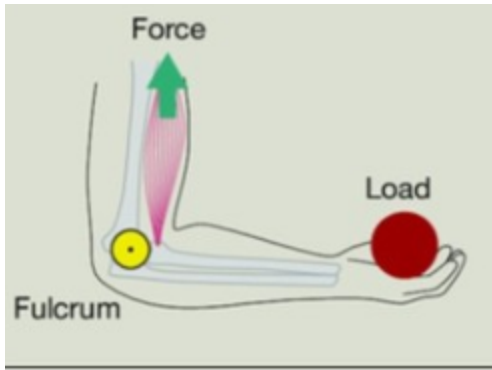


Question 3 14.5 pts

Some levers amplify motion and some amplify force.

Our forearm is a lever. The elbow is the fulcrum. This machine :

hint: not helping you. See my slides about levers and machines. you see the type and if amplifies motion or force.


☐

is not a machine

☐

amplifies motion

☐

I missed the class

☐

amplifies force

☒

Question 4 14.5 pts



What is true about this lever?

hint: see my slides.

☐

Its a second class level and amplifies effort

☐

Its a second class level and amplifies motion

☐

Its a third class class and amplifies motion

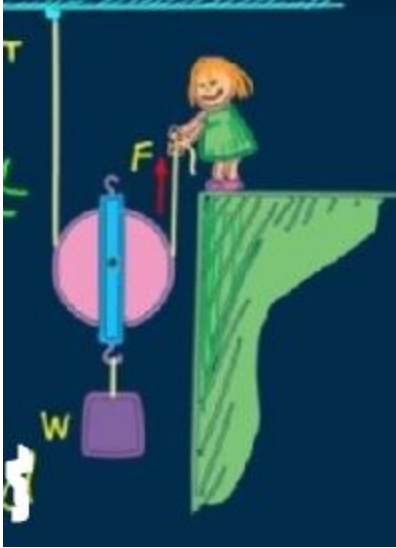


Its a first class lever and it amplifies effort



Question 5 14.5 pts

Here is little Suzie and her machine. not helping you



She is lifting a 10pounds weight (load). The load (10 pounds) moves by 2 yards. What is true.



her effort is 10 pounds, she pulls 4 yards. dah



I spaced out during class



her effort is 5 pounds, she pulls 2 yards



her effort is 5 pounds, she pulls 4 yards



her effort is 10 pounds, she pulls 4 yards



Question 6 15 pts

Use my slide about levers and machine. A class 3 lever .. What is true?



has the fulcrum between the effort and the load



has the load between the fulcrum and the effort



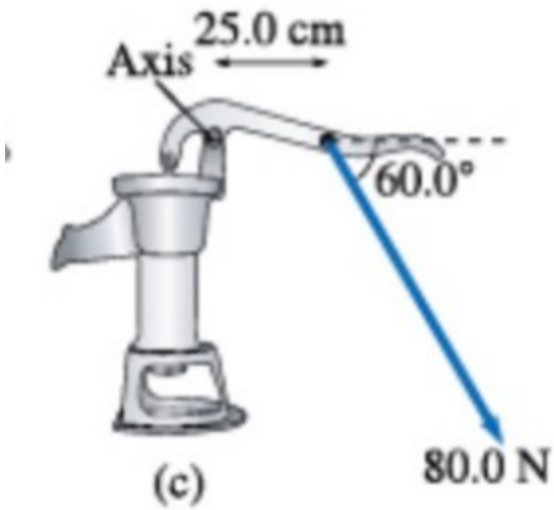
always amplifies force



has the effort between the fulcrum and the load



Question 7 14 pts



What is the torque produced by the 80N force (in magnitude). Shown here?

What units!



about 10 N.m



about 17N.m



about 20N..



No answer text provided.

Not saved

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